

Technical Report

**Edge Based Hybrid System Implementation
for Long Range Safety and Healthcare IoT
Applications**

Submitted By

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List of Abbreviations

Abbreviation	Full Form
CI	Computational Intelligence
ECG	Electrocardiogram
CNN	Convolutional Neural Network
1-D CNN	One-Dimensional Convolutional Neural Network
GRU	Gated Recurrent Unit
Bi-GRU	Bidirectional Gated Recurrent Unit
ANFIS	Adaptive Neuro-Fuzzy Inference System
FIS	Fuzzy Inference System
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
DL	Deep Learning
GA	Genetic Algorithm
FL	Federated Learning
RNN	Recurrent Neural Network
LoRa	Long Range

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1 Abstract

The integration of edge computing and artificial intelligence has redefined the efficiency and responsiveness of modern healthcare IoT systems. However, traditional cloud-dependent frameworks continue to face challenges such as high latency, limited bandwidth, and data privacy concerns. This paper presents an Edge-Based Hybrid System Implementation for Long-Range Safety and Healthcare IoT Applications [1] that leverages a combination of 1D Convolutional Neural Networks (1D-CNN), Gated Recurrent Units (GRU), Fuzzy Logic, and Genetic Algorithms (GA) to achieve intelligent and real-time ECG signal analysis directly at the edge.

The proposed framework employs a lightweight 1D-CNN-GRU hybrid model for feature extraction and pattern recognition, while a Fuzzy Inference System enhances interpretability and reduces uncertainty in classification. The Genetic Algorithm optimizes model hyperparameters, ensuring efficient performance under varying computational and energy constraints. An adaptive sampling mechanism adjusts the data acquisition rate based on signal variability and patient condition. During abnormal or high-risk conditions, the sampling rate is increased to capture finer signal details for accurate diagnosis.

By processing signals locally on edge devices, the system minimizes latency and bandwidth usage, achieving fast response times and robust operation in remote healthcare environments. Experimental evaluation demonstrates that the hybrid approach improves diagnostic accuracy, inference speed, and resource utilization compared to conventional cloud-based or single-model solutions. The design ensures energy efficiency, scalability, and privacy preservation, making it suitable for continuous long-range health monitoring. Future work will explore the integration of Federated Learning (FL) for collaborative edge intelligence, further enhancing data security and global model adaptability.

Keywords: Edge Computing, Healthcare IoT, ECG Signal Processing, 1D-CNN, GRU, Fuzzy Logic, Genetic Algorithm, Federated Learning, Router, Gateway, Real-Time Monitoring, Sampling Time.

2 Introduction

2.1 Background

Edge-based systems are essential in modern healthcare and safety applications where continuous monitoring and quick response are required. Transferring all sensor data to the cloud causes high latency, bandwidth overload, and energy inefficiency. To overcome these issues, computation is moved to the edge, enabling real-time processing near the data source.

This project implements an Edge-Based Hybrid System for real-time ECG analysis and health condition classification. An Router handles signal preprocessing using bandpass, notch, and wavelet filters to remove noise, while a Gateway executes advanced Computational Intelligence (CI) algorithms.

The hybrid CI model integrates a 1D-CNN for feature extraction, a GRU for analysis, and Fuzzy Logic for interpretable classification. A GA optimizes model parameters to enhance accuracy and efficiency. Additionally, an adaptive sampling mechanism dynamically adjusts sampling rates based on signal activity to balance energy use and response speed.

Overall, the system achieves low latency, efficient resource use, and reliable real-time monitoring suitable for long-range IoT healthcare applications.

2.2 Motivation

In long-range healthcare and safety systems, quick detection of critical conditions is crucial. However, depending entirely on cloud servers can be problematic in real-time environments. There is a need for a smart and efficient system capable of processing data and making accurate decisions directly at the edge.

There are limitations of static sampling and fixed resource allocation in existing systems. Constant high-rate sampling wastes power, while low-rate sampling may miss vital signal changes. The proposed system integrates an adaptive sampling technique that intelligently adjusts the sampling rate according to the patient's condition, ensuring accuracy.

GA, a deep learning model used for fine-tuning, is used to automatically optimize the parameters of the CNN-GRU-Fuzzy hybrid model, allowing the system to achieve high accuracy while maintaining low latency and minimal power consumption.

Therefore, the project aims to develop a computationally intelligent edge system that performs real-time ECG analysis, adapts resource usage, and ensures reliable operation in long-range healthcare monitoring applications.

2.3 Problem Definitions

The aim of this research is to address the challenges in real-time health monitoring on edge devices, focusing on reducing latency, improving classification accuracy, and handling uncertainty in physiological data. The proposed solution leverages a hybrid computational intelligence model combining 1D-CNN, GRU, and Fuzzy Logic for efficient decision-making. To develop a hybrid computational intelligence model that integrates 1D-CNN for feature extraction, GRU for pattern recognition, and Fuzzy Logic for interpreting uncertain physiological signals, enabling robust real-time health monitoring on resource-constrained edge devices.

2.3.1 Specific Goals

- Minimize latency and bandwidth usage in IoT sensor networks.
- Enhance the accuracy and reliability of health state classification.
- Incorporate fuzzy reasoning to handle uncertainty in biomedical signals.
- Optimize computational efficiency for low-power edge devices.

2.3.2 Input and Output

- **Input:** Biomedical signals (ECG) acquired from wearable IoT sensors.
- **Processing:**
 1. CNN extracts features from input signals.
 2. GRU captures dependencies and sequential patterns.
 3. Fuzzy Logic interprets model outputs under uncertain conditions to generate actionable health insights.
- **Output:** Classified health state (e.g., normal or classes of abnormalities) for timely intervention.

2.3.3 System Constraints

- Edge devices have limited memory and processing power.
- Communication occurs over low-bandwidth networks (LoRa, WiFi, BLE).
- Real-time response is critical, requiring decisions within milliseconds.

3 Literature Survey

[2] Kiranyaz, Ince, et al. proposed a model that focuses on lightweight 1D-CNN architecture designed for efficient ECG classification on resource-constrained edge devices, eliminating the need for cloud-based inference. It incorporates patient-specific training to enhance sensitivity toward individual ECG variations, achieving robust performance in distinguishing normal and abnormal heartbeats. Experiments on standard ECG datasets validate its high accuracy and low-latency operation, making it suitable for wearable health monitoring applications. However, the model primarily extracts spatial features and is limited to single-lead ECG signals, restricting its ability to capture long-term temporal dependencies and broader clinical applicability.

The study addresses key challenges such as designing a compact real-time model for low-power devices, adapting to patient-specific ECG patterns, and ensuring reliability under noisy or dynamic conditions. Research gaps include improving model generalization across heterogeneous datasets, integrating long-term temporal dependencies in a lightweight manner, and extending applicability to multi-lead ECG data for comprehensive cardiac monitoring.

[3] Mahel et al. proposed a hybrid CNN-GRU model with attention for arrhythmia detection, integrating spatial and temporal feature extraction. CNN layers capture local ECG patterns, while GRU layers model sequential dependencies. The attention mechanism highlights critical signal segments, enhancing interpretability. Validated on public ECG datasets, the model outperforms standalone CNN or GRU approaches and provides explainable insights into which parts of the ECG drive predictions. While offline evaluation demonstrates high classification accuracy, the study does not explore energy-efficient or real-time deployment strategies.

Key challenges include effectively combining CNN and GRU layers, designing attention mechanisms for improved interpretability, and managing the trade-off between accuracy, transparency, and computational complexity. Research gaps involve optimizing attention mechanisms for faster inference on low-power hardware, validating the model on larger and more diverse ECG datasets for scalability, and developing strategies for real-time deployment on resource-constrained edge devices.

[4] Reshad et al. introduced a stacked CNN with bidirectional GRU and attention for arrhythmia detection, combining hierarchical spatial feature extraction with temporal modeling of past and future dependencies. The attention mechanism highlights important ECG segments, improving interpretability and sen-

sitivity. Evaluated on multiple datasets, the model achieves high accuracy and effectively captures long-term temporal dependencies, making it suitable for detecting complex arrhythmia patterns. However, evaluation is primarily offline, and the model's high computational requirements limit its assessment on real-time or resource-constrained platforms.

The main challenges involve capturing long-range temporal dependencies with Bi-GRUs, integrating CNN, Bi-GRU, and attention modules for effective feature fusion, and maintaining stable training with efficient computation in a deep hybrid network. Research gaps include designing lightweight Bi-GRU and attention architectures for edge deployment, testing generalizability across diverse patient populations, and optimizing input sequence lengths to balance performance and computational efficiency.

[5] Keikhosrokiani et al. proposed system that integrates CNN layers for spatial feature extraction, Bi-GRU for temporal dynamics, and ANFIS for fuzzy reasoning to handle uncertainty in biomedical signals. This hybrid approach demonstrates robust performance on noisy data, outperforming standalone CNN or RNN models, while improving interpretability and reliability through explainable hybrid learning. However, the model is computationally intensive and primarily evaluated offline, limiting its applicability in real-time scenarios.

Key challenges include effectively integrating CNN–BiGRU–ANFIS components, managing noisy biomedical signals while preserving feature interpretability, and optimizing parameters between neural and fuzzy modules for balanced accuracy and efficiency. Research gaps involve adapting this hybrid framework to other biomedical signals like ECG, reducing computational complexity for edge deployment, and validating performance on larger, more diverse datasets to ensure robustness and generalizability.

4 Problem Formulation

Modern healthcare IoT systems rely heavily on cloud computing for continuous patient monitoring and analysis. While effective in theory, this dependency introduces challenges such as high latency, bandwidth overload, data privacy concerns, and inefficient energy use—all of which can delay critical decision-making. In real-time safety and healthcare applications, such delays or data losses may have serious consequences.

The main problem addressed in this work is the design of a computationally intelligent edge-based system capable of performing real-time ECG signal processing and classification locally, with minimal reliance on cloud computation. The system must efficiently manage resource allocation, sampling rate, and data transmission while maintaining high accuracy and low power consumption suitable for long-range healthcare environments.

To achieve this, the project integrates CI techniques namely 1D-CNN for feature extraction, GRU for sequence learning, and Fuzzy Logic for interpretable classification. A GA is employed to optimize model parameters for improved performance under limited computational resources. The system also includes an adaptive sampling mechanism that dynamically adjusts the sampling time based on ECG signal activity, balancing accuracy and energy efficiency.

4.1 System Overview

The proposed Edge-Based Hybrid System operates across three layers—Input Layer, Edge Layer, and Cloud Layer—each handling a specific stage of data flow and processing to achieve intelligent, real-time ECG analysis.

Input Layer

The Input Layer is responsible for data collection through ECG sensors attached to the subject. These sensors continuously capture analog physiological signals and transmit them to the edge devices. No computation is performed here; the focus is solely on accurate and continuous signal acquisition to ensure reliable input for the next stage.

Edge Layer

The Edge Layer performs both signal preprocessing and intelligent analysis, making it the core of the system. It consists of two key hardware components: Router: Handles signal preprocessing using bandpass, notch, and wavelet filters to remove noise and interference. This ensures that only clean, relevant

data are passed to the next stage while conserving energy and reducing communication load. Gateway: Executes the Computational Intelligence model, integrating 1D-CNN, GRU, and Fuzzy Logic. The GA fine-tunes model parameters to achieve optimal accuracy and processing efficiency. An adaptive sampling mechanism dynamically modifies the sampling rate based on signal variation—reducing power consumption during stable conditions and increasing responsiveness during irregular heart activity.

This layer performs all real-time processing and decision-making at the edge, minimizing the need for cloud dependence.

Cloud Layer

The Cloud Layer is dedicated to data storage and management. Processed outputs and classified results from the edge are securely stored in the cloud for long-term record keeping, trend analysis, and remote access by medical personnel. No real-time computation is performed here, ensuring low latency and improved privacy while retaining historical access to health data.

5 Proposed Architecture

The Proposed Architecture, Figure 1 and Figure 2, provides the practical implementation framework for the formulated problem. It follows a three-tier structure—Input, Edge, and Cloud—ensuring optimized processing, minimal delay, and reliable long-range data handling.

The Input Layer serves as the sensing interface, collecting ECG data in real time. The Edge Layer acts as the computational core, where the Router performs preprocessing and the Gateway executes the hybrid CI model (1D-CNN + GRU + Fuzzy Logic). The GA refines the model parameters to enhance classification accuracy while maintaining computational efficiency. The adaptive sampling module dynamically manages signal acquisition frequency to optimize energy use and responsiveness.

Finally, the Cloud Layer stores the processed results for long-term analysis, reporting, and system updates. This division of tasks ensures that data are processed locally for faster response, while storage remain securely managed in the cloud.

Overall, the proposed architecture provides a scalable, energy-efficient, and intelligent edge computing framework capable of reliable real-time ECG classification and long-range health monitoring.

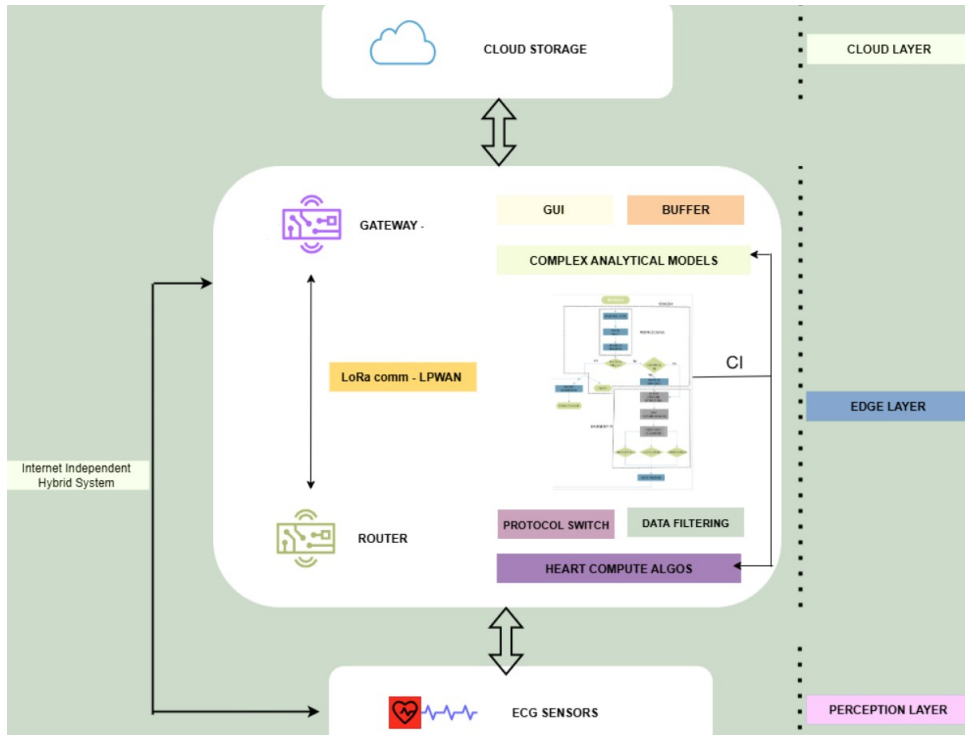


Figure 1: Architecture Diagram - both CI and Edge

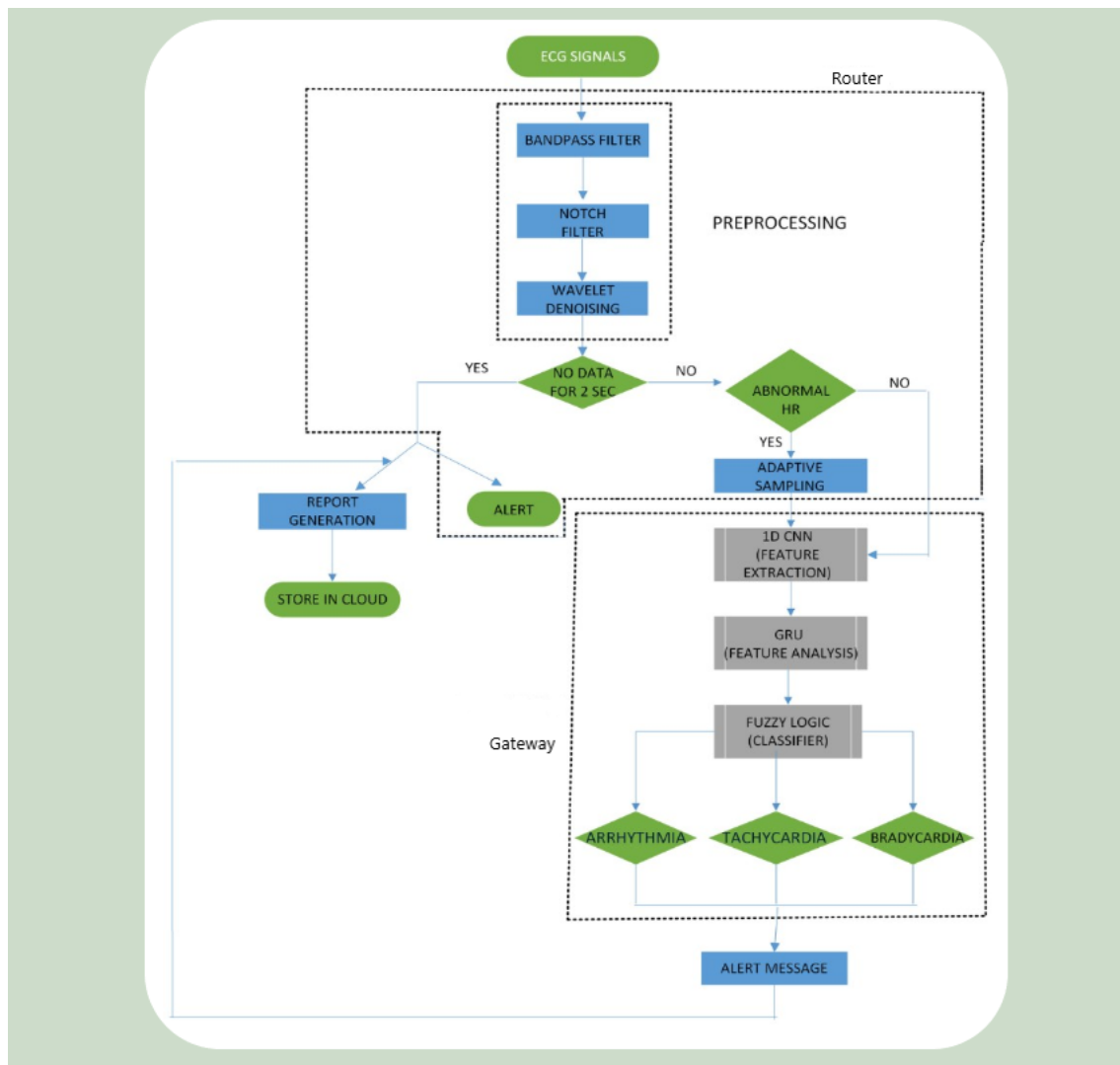


Figure 2: Architecture Diagram - only CI

6 Methodology

The proposed system follows a structured five-stage methodology to achieve accurate and real-time ECG signal classification on edge devices. Each stage is optimized for low latency, computational efficiency, and clinical reliability in long-range healthcare IoT applications. The architecture integrates traditional signal processing, deep learning, and fuzzy reasoning for robust and interpretable health assessment.

6.1 Preprocessing

Raw ECG signals are first processed to remove noise and interference introduced during acquisition. A bandpass filter(0.5–40 Hz) is used to eliminate baseline wander and high-frequency noise. A notch filter centered at 50/60 Hz suppresses power-line interference. Wavelet-based denoising further refines the signal by reducing motion artifacts and transient disturbances without distorting clinically relevant features.

This multi-stage preprocessing ensures high signal quality, essential for reliable feature extraction in subsequent stages. All these operations are implemented on the Router , allowing efficient on-board signal conditioning before transmission to the edge node.

6.2 Adaptive Sampling

An adaptive sampling mechanism dynamically adjusts the sampling rate based on signal behavior. During stable cardiac activity, a lower sampling rate minimizes data load and power consumption. When irregularities or abnormalities are detected, the system automatically increases the sampling rate to capture finer ECG details. This adaptive strategy enhances both energy efficiency and diagnostic precision, making the system suitable for continuous long-term monitoring.

6.3 One-Dimensional Convolutional Neural Network (1D-CNN)

A neural network that applies one-dimensional convolutional filters to sequential data, extracting local patterns and features from signals such as ECG. The preprocessed ECG data is fed into a 1D Convolutional Neural Network for local feature extraction. Convolutional layers detect morphological patterns such as P-waves, QRS complexes, and T-waves. The CNN effectively captures dependencies and local signal variations, forming a compact representation of the cardiac cycle. This module performs the initial interpretation of heart signal morphology and identifies potential irregularities.

6.4 Gated Recurrent Unit (GRU) Network

A type of recurrent neural network with gating mechanisms that efficiently captures temporal dependencies in sequential data while mitigating vanishing gradient problems. The extracted CNN features are passed into a GRU network to analyze dependencies between consecutive cardiac cycles. The GRU efficiently models sequential ECG dynamics to distinguish normal from abnormal rhythms. Its gating mechanism allows selective information retention, reducing memory usage compared to traditional RNNs or LSTMs. The GRU layer acts as the final deep learning stage, performing sequence-level evaluation and generating a preliminary classification output.

6.5 Fuzzy Logic Decision Layer

A system that applies fuzzy reasoning to translate numerical inputs into linguistic outputs, handling uncertainty and imprecision in decision-making. The final decision-making process is governed by a Fuzzy Inference System (FIS). Inputs from the CNN and GRU outputs are combined with expert-defined fuzzy rules. The FIS converts numerical probabilities into interpretable linguistic categories such as Normal, Arrhythmia, Bradycardia or Tachycardia. Membership functions and fuzzy reasoning handle signal uncertainty, noise, and ambiguous conditions that may not be captured by neural networks alone.

The proposed edge-based hybrid pipeline leverages signal processing for data refinement, adaptive sampling for efficiency, 1D-CNN and GRU for feature learning, and fuzzy logic for interpretable decision-making. Together, these modules create an intelligent, low-power, and real-time ECG monitoring system optimized for long-range healthcare IoT applications.

7 Results

The Computational Intelligence (CI) model was trained and validated using the MIT-BIH Arrhythmia dataset from PhysioNet [6], which contains ECG recordings from multiple patients with various cardiac conditions. The dataset was processed to include four ECG classes: Normal, Bradycardia, Tachycardia, and Arrhythmia.

7.1 Preprocessing

Before training, the dataset underwent several preprocessing steps to improve data quality and training efficiency. The raw ECG signals were segmented into 250-sample beats around each annotated R-peak. Only complete beats were considered. The segments were labeled according to their respective classes, encoded numerically, and converted into one-hot vectors for multi-class classification. The dataset was then reshaped for Conv1D input and split into training (80%) and test (20%) sets, ensuring stratified class distribution.

This preprocessing ensured that the input signals were consistent, reduced the model's computational load, and improved convergence speed. These same filtering and segmentation techniques were later replicated on the Router to maintain consistency between the trained model and the real-time input data.

7.2 Model Training

A hybrid 1D-CNN + GRU model was used for ECG classification. The CNN layers extracted spatial features from the ECG beats, while the GRU layer learned temporal dependencies between waveform points. A Dropout layer was added for regularization, and the output layer used softmax activation for 4-class classification.

The model was trained for 10 epochs using the Adam optimizer and categorical cross-entropy loss with a batch size of 64. The validation accuracy steadily improved over epochs, indicating stable learning and good generalization performance.

7.3 Fuzzy Logic Interpretability

To provide interpretable diagnostic outputs, a Fuzzy Logic system was applied on top of the CNN-GRU model's softmax probabilities. The system converts model confidence levels into human-understandable categories such as:

- Heart Rate = 80 bpm, Confidence = 0.95 $\rightarrow$ Class 0 (Normal)

- Heart Rate = 150 bpm, Confidence = 0.90 → Class 2 (Tachycardia)

This ensures that the edge-deployed model can provide actionable insights in real-time, bridging the gap between automated classification and clinical interpretability.

7.4 Performance Evaluation

The model's performance was evaluated using standard metrics such as accuracy (Figure 3) and loss (Figure 4), mentioned in Table 1. This demonstrates the model's reliability for real-time ECG classification tasks.

- class 0 - Normal
- class 1 - Bradycardia
- class 2 - Tachycardia
- class 3 - Arrhythmia

Table 1: Performance Evaluation

Evaluation Metric	Value
Accuracy	visible increase (from 0.9601 to 0.9850)
Loss	visible decrease (from 0.2219 to 0.0694)



Figure 3: Accuracy

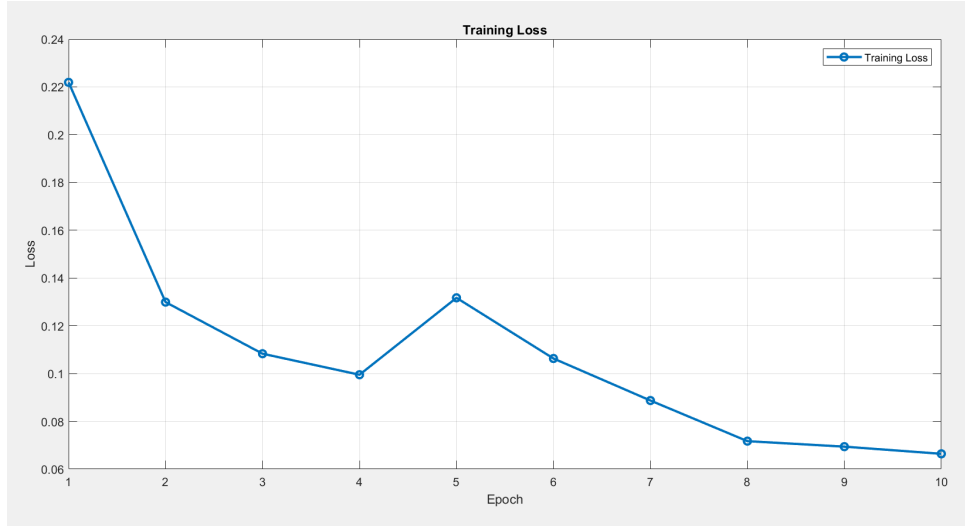


Figure 4: Loss

7.5 Model Deployment

After training, the model was saved as `ecg_classifier_model.h5`, along with the Fuzzy Logic mapping rules. This modular saving approach allows the model to be reloaded for inference or fine-tuning.

The trained model, combined with the Fuzzy Logic system, can be deployed on edge hardware to classify preprocessed ECG beats in real-time, maintaining high accuracy and providing interpretable outputs for clinical decision support.

7.6 Summary

The results demonstrate that the proposed CI-based edge system can effectively classify ECG signals with high accuracy while remaining computationally efficient. The preprocessing pipeline, hybrid 1D-CNN-GRU-Fuzzy architecture, and GA optimization collectively contribute to a robust, low-latency model suitable for real-time, long-range healthcare IoT applications.

8 Conclusion

This work presents a hybrid edge-based system integrating 1D-CNN, GRU, and Fuzzy Logic for real-time ECG analysis in healthcare IoT applications. The architecture effectively combines traditional signal preprocessing with adaptive sampling to ensure high-quality input and efficient resource utilization on edge devices. The 1D-CNN and GRU jointly enable accurate feature extraction, while the fuzzy inference layer enhances interpretability and robustness in decision-making under uncertain signal conditions.

By performing computation locally at the edge, the system achieves reduced latency, lower bandwidth usage, and improved data privacy compared to conventional cloud-based approaches. The integration of a GA for parameter optimization further refines model performance and adaptability across varying conditions.

Overall, the proposed framework demonstrates a scalable, efficient, and intelligent solution for continuous, long-range health monitoring. Future extensions include the incorporation of FL to enable collaborative training across distributed healthcare nodes, enhancing model generalization and privacy preservation.

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